

Demonstration of Proposed Features

Date	27 June 2026
Team ID	SWUID2026217448
Project Name	A Comprehensive Measure of Well-Being (HDI Predictor)
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	HDI Prediction (Very High/High/Medium/Low)	Predicts a country's Human Development Index tier based on life expectancy, schooling, and GNI per capita	Implemented	Yes	Core feature, working as expected
2	Data Preprocessing & Label Encoding	Cleans dataset and encodes categorical HDI tiers for model training	Implemented	Yes	Handled missing values and outliers
3	Machine Learning Model (Classification)	Scikit-learn based model trained on HDI indicators to classify development level	Implemented	Yes	Achieved good accuracy on test data
4	Web Interface (Flask + HTML)	User-friendly form to input indicators and view predicted HDI category	Implemented	Yes	Built with Flask, index.html and result page
5	Model Deployment (.pkl integration)	Trained model and label encoder saved and loaded into the Flask app for real-time prediction	Implemented	Yes	Uses HDI.pkl and label_encoder.pkl
6	Scenario-based Testing	Validated predictions using Very High, Medium and Low development scenarios	Implemented	Yes	Verified against the three documented scenarios

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6

Overall Implementation Rate (%)	100%
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